

UNIVERSITY OF HAMBURG

MASTER THESIS

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UNIVERSITY OF HAMBURG

Abstract

Institute of Psychology

Department of Psychology with focus on Quantitative Methods

Master of Science

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by Lona Frießner

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Chapter 1

Introduction

Chapter 2

Theory

Chapter 3

Multilevel models

Example

Imagine that you are a researcher interested in the musical self-concept of young adults. You want to examine whether there is a connection to weekly duration of practice on their instrument.

So you gather a sample. You manage to find 15 youth ensembles that contribute to your study. For simplicity, let's consider that there are 10 players in each ensemble. Each player gets a questionnaire with items about their musical self-concept and their weekly practice duration.

3.1 The idea behind multilevel models

Populations like this, where units of interest (e.g., music players, students, employees) are grouped into higher-level units (e.g., ensembles, classes, firms), are common in the human social life and therefore in psychological studies.

The resulting data is called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7).

From a sampling perspective, a so-called *multistage sampling process* (sample groups first, then individuals inside these groups¹) is not independent: After selecting a group, you will try to sample all of the members of this group, making it more likely to select one person in this group over others in non-selected groups (Snijders & Bosker, 2012, p. 7). Also, individuals inside the same group typically share contextual influences (e.g., the same ensemble teacher, the same class room) and thus will give more similar responses in surveys and behavior. As a consequence, the basic assumption of independence in traditional linear modeling is violated, because error terms will be correlated (Raudenbush & Bryk, 2010, p. xx).

One well-suited and popular possibility to account for this fact is the use of *multilevel models* (also called *hierarchical linear models* or *mixed models*).

A multilevel model (MLM) extends regular linear regression by adding *random effects*

3.2 The random intercept model with context effects

¹For simplicity, this thesis does not address three (or higher)-level designs.

Chapter 4

Small sample sizes in multilevel models

Auch Bayesian, Raudenbush S. 399

Chapter 5

Missing data in multilevel models

5.1 Methods for handling missing data

5.1.1 Listwise Deletion

5.1.2 Full Information Maximum Likelihood

5.1.3 Bayesian Model Estimation

5.1.4 Multiple Imputation

Chapter 6

Methods

A simulation study was conducted in order to compare the performance of methods for missing data handling in small-sample multilevel random intercept models.

The idea behind a simulation study is to carry out a computer experiment (Morris et al., 2019), where data with known characteristics is simulated and subsequently analyzed with the methods of interest. The analysis results can be compared to the truth, allowing for assessment of the performance of the methods (Siepe et al., 2024).

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

6.1 Data generation mechanism

6.1.1 Data-generating model

The data-generating model is a two-level random intercept model:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + \gamma_{02}W_j + u_{0j} + e_{ij} \quad (6.1)$$

Y_{ij} , X_{ij} and W_j are parameterized to have a mean of zero and a variance of 1 on the population level.

First, between- and within-components of X_{ij} were simulated, independent of each other and with variances chosen to achieve a prespecified intraclass coefficient (see Section 6.1.3):

$$\begin{aligned} X_{ij} &= u_{Xj} + e_{Xij}, \\ u_{Xj} &\sim N(0, \tau_x^2), \quad e_{Xij} \sim N(0, \sigma_x^2) \end{aligned}$$

with $\tau_x^2 = ICC_x$ and $\sigma_x^2 = 1 - ICC_x$, respectively.

X_{ij} was then group-mean-centered.

The group-level covariate W_j was constructed with a linear regression equation to have a prespecified correlation with $\bar{X}_{\cdot j}$ ($\rho_{\bar{X}W}$):

$$W_j = a\bar{X}_{\cdot j} + e_{Wj}, \quad e_{Wj} \sim N(0, \sigma_w^2)$$

The residual variance σ_w^2 and the regression coefficient a are fully determined by the restriction that $Var(W_j) = 1$:

$$\begin{aligned} \sigma_w^2 &= 1 - \rho_{\bar{X}W}^2, \\ a &= \frac{\rho_{\bar{X}W}}{\sqrt{\tau_x^2 + \frac{\sigma_x^2}{n}}} \end{aligned}$$

TABLE 6.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

Data conditions	
N_1	10
N_2	15 / 30 / 60
ICC of X and Y	.1 / .3
Correlation $\bar{X}W$	.3
Model parameters	
Effect of $(X_{ij} - \bar{X}_{.j})$	.2
Effect of $\bar{X}_{.j}$	0 / .4
Effect of W_j	0
Missing values	
Pattern	$X \sim W$
Mechanism	MCAR / MAR
Proportion	.3

Finally, the random effects on the group and the individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \quad e_{ij} \sim N(0, \sigma^2)$$

and independent of each other.

Individual and group residual variances of Y_{ij} are determined by the constraints of a prespecified residual ICC of Y_{ij} , $Var(Y_{ij}) = 1$ and $Var(X_{ij}) = 1$ ¹:

$$\begin{aligned} \psi^2 &= ICC_{Y|X,W} \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n_j})) \\ \sigma^2 &= (1 - ICC_{Y|X,W}) \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n_j})) \end{aligned}$$

The full derivation of these equations is provided in [?@sec-appA](#).

6.1.2 Missing data generation

After simulation of the complete dataset, values of the X -variable have been deleted to simulate missingness. Following an approach by Grund (2023, unpublished HTML-Document), a ‘missingness tendency’ r was simulated for each X -value according to this equation:

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \quad e_{ij} \sim N(0, 1 - \beta^2)$$

r_{ij} is a normally distributed, continuous variable. When $r_{ij} > 0$, the ij -person’s X -value was deleted.

The intercept α controls the proportion of missingness in X , whereas the slope β determines the strength of the missing-at-random mechanism. When $\beta = 0$, missingness is completely at random (MCAR). With increasing β , the missingness mechanism becomes more dependent on the W -values; with $\beta = 1$, missingness would be completely deterministic.

6.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table [6.1](#).

¹as well as setting γ_{02} to be zero across all conditions, see Section [6.1.3](#)

The number of Level-2 units was manipulated across conditions ($N_2 = 15, 30, 60$), while the number of Level-1 observations was kept constant ($N_1 = 10$). The between-cluster effect of $\bar{X}_{\cdot j}$ was varied ($\gamma_{01} = 0, .40$), whereas the within-cluster effect of $(X_{ij} - \bar{X}_{\cdot j})$ was held constant at .20 and the effect of W_j was set to zero. This results in conditions with either a negative and a positive context effect (i.e., $\gamma_{10} - \gamma_{01}$) of X in the population model. The ICC of X and the residual ICC of Y were varied ($\rho_X = \rho_{Y|X} = .10, .30$). Missing data were generated on X with a varying missingness mechanism (MCAR or MAR with a strength of $\beta = .30$), with a constant missingness rate of 30%. The factors were combined fully factorial, creating $3 \times 2 \times 2 \times 2 = 24$ conditions.

All in all, these settings reflect a number of the typical applications in cross-sectional and longitudinal applied research (e.g., smaller and larger intraclass correlation coefficients (ICC) and level-two sample sizes). They also create sufficient methodological variation to evaluate the properties of the methods of interest (e.g., null versus substantial effects).

6.2 Estimands of the focal analysis model

The focal analysis model is

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + u_{0j} + e_{ij} \quad (6.2)$$

The primary target is the Level-2-effect, γ_{01} . Additionally, the Level-1-effect γ_{10} was evaluated.

6.3 Methods of Missing Data Handling

Four methods will be compared, alongside the benchmark analysis of Complete Data (CD).

6.3.1 Listwise Deletion (LD)

Listwise Deletion was used as a standard reference method for missing data handling. Cases with missing X -values were excluded prior to analysis. The group means $\bar{X}_{\cdot j}$ were then calculated only from the remaining observations. Then, group-mean centering was applied.

6.3.2 Multiple Imputation (MI) with standard (MI-R) and adjusted (MI-a) degrees of freedom

For both multiple imputation methods, a joint modeling approach was employed. Although only values in X were missing, the multivariate specification preserves the multilevel structure during imputation:

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix}$$

The random effects $(u_{Yj}, u_{Xj}, u_{Wj})^\top$ and the residuals $(e_{Yij}, e_{Xij})^\top$ follow multivariate

normal distributions with mean zero and covariance matrices $\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}$

and $\Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}$

W_j serves as an auxiliary variable to explain missingness.

Imputation was executed with the R package `jomo` (Quartagno & Carpenter, 2023) with 2000 burn-in-iterations and 3000 post-burn-in iterations, saving every 300th imputation (10 imputations).

For MI-R, degrees of freedom for calculating the test statistic were calculated with the software standard (Rubin's rules, assuming infinite degrees of freedom in complete data).

For MI-a, degrees of freedom were calculated with $df_{\gamma_{10}} = M - r - 1$ (M : total number of Level-1 units; r : total number of predictors, in the present case 2) and $df_{\gamma_{01}} = N_2 - q - 1$ (N_2 : number of Level-2 units; q : number of Level-2 predictors, here: 1)

6.3.3 Fully Bayesian Analysis

A factored regression specification was used within the Bayesian framework to allow the inclusion of the auxiliary variable W while leaving the focal analysis model unchanged:

$$g(y, x, w; \gamma) = g_w(w|y, x; \gamma_w) \times g_y(y|x; \gamma_y) \times g_x(x; \gamma_x) \quad (6.3)$$

Here, g_w is the conditional distribution of the auxiliary variable, while g_y corresponds to the substantive analysis model.

The burn-in period was set to 1000 iterations, followed by 3000 post-burn-in iterations. The model was estimated using the package `mdmb` (Robitzsch & Luedtke, 2024) with its default noninformative priors.

6.3.4 Extracted quantities

For complete data, listwise deletion and both multiple imputation methods, point estimates of γ_{10} and γ_{01} as well as their standard errors and 95% - confidence intervals were extracted.

For the fully Bayesian method, the posterior means of γ_{10} and γ_{01} were used as point estimates along with posterior standard deviations and 95% credible intervals. For comparability with the frequentist approaches and in line with common simulation practice, these posterior summaries were treated analogously to frequentist estimates and intervals (Morris et al., 2019).

6.4 Performance measures

Three performance measures were used for each parameter (γ_{10} and γ_{01}):

1. Bias of point estimates, defined as the difference between the parameter's mean estimated and true value.
2. Coverage of confidence/credible intervals, defined as the proportion of simulation repetitions that captured the true parameter value inside the CI.
3. Standard deviation of the estimations across simulation repetitions.

Monte Carlo uncertainty of the estimated performance measures was calculated using Monte Carlo Standard Errors (MCSE), using the formulas provided in Siepe et al. (2024).

For each condition, 1000 simulation replications were used, based on comparable simulation studies (Enders et al., 2016; Grund et al., 2018) and ensuring a sufficiently small MCSE of $< 1\%$ for coverage [[^]methods-2: MCSE calculation for bias and standard deviation are based on the sample variance of the estimates and cannot be pre-calculated.].

The simulation was conducted in RStudio with the aim of full reproducibility, following guidelines of the repro book (ZITIEREN), e.g. by setting a seed for simulation and using reproducible package environments. The code and all relevant information for reproduction can be found in the digital version of the master thesis. (LINK ZU GITHUB)

INSGESAMT NOCH MEHR BESCHREIBEN? ALSO AUCH, WELCHE GEDANKEN ICH MIR ZUR AUSWERTUNG GEMACHT HATTE.

Chapter 7

Results

The complete results for the parameters of interest are provided in tables in [?@sec-appB](#).

7.1 Bias of point estimation

In line with expectations, no substantial bias was observed for the individual-level effect γ_{10} . The findings for γ_{01} are depicted in Figure 7.1 For γ_{01} , listwise deletion produced a substantial bias especially in low-ICC conditions, with positive bias on the null effect and negative bias on the positive between-group effect. With increasing sample size, this bias substantiates across all conditions and gets bigger in MAR conditions (see .

Surprisingly, bias in multiple imputation was even higher than LD-bias in conditions with low ICC and positive effect, while the Bayes method only led to bias in the toughest conditions (smallest sample size, low ICC and positive γ_{01} -effect}.

7.2 Coverage of confidence/credible intervals

7.3 Standard deviation of point estimates

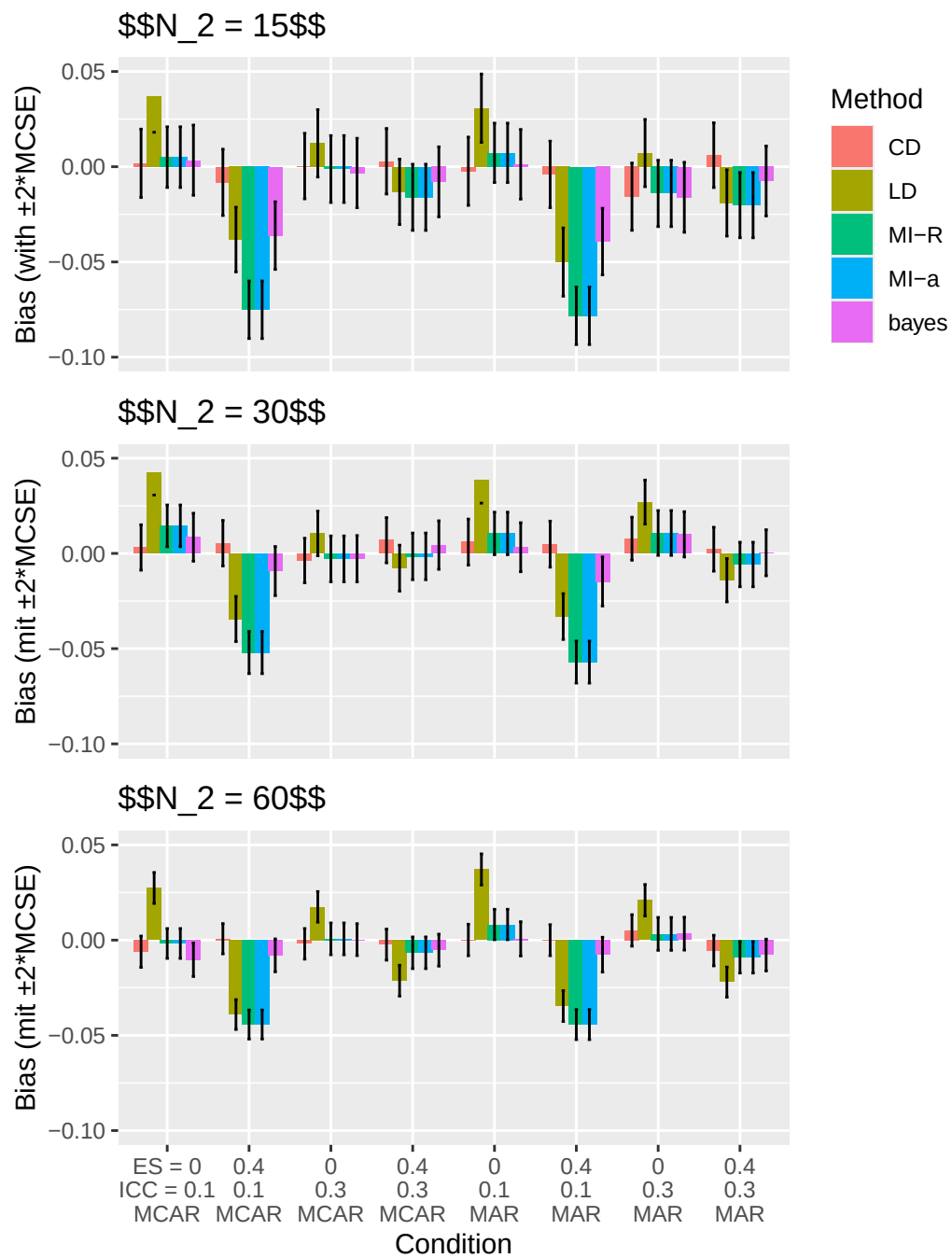


FIGURE 7.1

Chapter 8

Discussion

MI Diskussion Priors ICC .4 gewinnt doppelt:

Prior passt besser

mehr echte Information (weil weniger within-Varianz)

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